2018-Ph-H1-Q3

Section: Particles and Waves

Topic: The Inverse Square Law

A radiation detector measures a count rate of 160 counts/s at a distance of 0.50 m from a radioactive source.

What is the count rate at a distance of 1.0 m?

Solution:

Use the inverse square law:

$$I \propto \frac{1}{r^2}$$

Doubling the distance reduces intensity by a factor of $2^2 = 4$.

$$\text{New count rate} = \frac{160}{4} = 40\,\text{counts/s}$$

Answer: 40 counts/s

Revision Tip:

In inverse square law questions, always square the distance ratio. Double the distance means one-quarter the intensity.